

HSE Intelligence Platform

Mobile Application — Role Workflow Guide

Single APK · 4 Roles · Role-Based Navigation

Worker	■	Report incidents, near misses, complete tasks, view training
Supervisor	■	Submit checklists, report events, update CAPA progress
HSE Manager	■■	Approve permits, validate checklists, conduct safety walks, assign CAPAs
Auditor	■	Read-only access to all reports and compliance data

WORKER WORKFLOW

■ Worker

Tabs:

Dashboard	Tasks	Training	Profile
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Login Flow

Step 1: Login

- Enter Employee ID + PIN
 - Backend validates credentials
 - JWT token issued with role='Worker'
 - App navigates to Worker tabs
- API: POST /api/v1/auth/login

Dashboard

Step 2: View Dashboard

- See assigned site, zone, shift schedule
- Quick action buttons visible: Report Incident, Report Near Miss

Report Incident

Step 3: Report Incident

- Tap 'Report Incident' button
 - Fill form: incident type, severity, description, location
 - Select hazard category
 - Submit form
- API: POST /api/v1/incidents

Backend: Creates incident record, auto-flags CAPA generation if severity is Lost Time or Critical.

Report Near Miss

Step 4: Report Near Miss

- Tap 'Report Near Miss' button
 - Fill: description, potential consequence, underlying cause
 - Select associated hazard
 - Submit
- API: POST /api/v1/near-misses

Tasks

Step 5: View & Perform Tasks

- See list of assigned tasks for today
- Tap task → PerformTask screen
- Complete task actions
- Mark as complete

Training

Step 6: Safety Training

- View assigned training modules
 - Tap module → SafetyTrainingDetail screen
 - Complete quiz or mark attendance
- API: GET /api/v1/training-programs

Profile

Step 7: Profile & Settings

- View name, role, site, department
- Change password
- Logout

SUPERVISOR WORKFLOW

■ Supervisor

Tabs:

Dashboard	Tasks	Checklist	Reports	Profile
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Checklist Submission (Primary Workflow)

Step 1: Select Checklist Type

- Daily Safety Walk
- Hot Work Pre-Task
- Working at Height
- End of Shift Handover

Step 2: Create Draft

- Select site, zone, shift
- Set checklist date
- Tap 'Create Draft'

API: POST /api/v1/checklists/submissions

Step 3: Fill Checklist Items

- Each item: Yes/No/NA response
- Add remarks for non-compliant items
- All items visible with section headers

Step 4: Save Draft (optional)

- Save progress without submitting
- Can return later to complete

API: PUT /api/v1/checklists/submissions/{uuid}/items

Step 5: Submit for Validation

- Tap 'Submit' button
- Status changes: draft → submitted
- Goes to HSE Manager queue for validation

API: POST /api/v1/checklists/submissions/{uuid}/submit

After submission: HSE Manager receives notification, reviews items, and approves/rejects.

Report Incident / Near Miss

Step 6: Report Events

- Same flow as Worker (Report Incident / Near Miss)
- Supervisor can also see team's reported incidents in Reports tab

CAPA Action Update

Step 7: Update CAPA Progress

- View assigned CAPA actions in Tasks tab
- Update status: Open → In Progress → request close
- Add progress notes

API: PATCH /api/v1/capa-actions/{id}

Reports

Step 8: View Reports

- Submitted checklists history with status (approved/rejectedd)
- Incidents reported by team
- Near misses from shift

HSE MANAGER WORKFLOW

■ ■ HSE Manager

Tabs:

Dashboard	Permits	Incidents	Checklist	Profile
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Permit to Work Management

Step 1: View Permit Requests

- List of all permit requests (Hot Work, Confined Space, WAH, etc.)
- Filter by status: Pending, Active, Closed

Step 2: Approve/Reject Permit

- Tap permit → Review: workers, location, duration, risk level
- Approve → status becomes 'Active'
- Reject → with rejection notes

API: PATCH /api/v1/permits-to-work/{id}

Step 3: Create New Permit

- Fill: permit type, location, work description, duration
- Assign issuer and approver
- Set validity start/end times

API: POST /api/v1/permits-to-work

Checklist Validation

Step 4: Review Submitted Checklists

- View queue of submitted checklists (from Supervisors)
- Tap checklist → See all item responses and remarks

Step 5: Validate

- Approve: all items satisfactory
- Reject: non-compliant items found, add notes
- Status: submitted → approved/rejected

API: POST /api/v1/checklists/submissions/{uuid}/validate

Incident Management & CAPA

Step 6: View All Incidents

- List of all incidents with severity badges
- Tap → full detail: timeline, root cause, CAPA actions

Step 7: Create CAPA Action

- From incident detail → 'Create CAPA'
- Fill: action type (Corrective/Preventive), description
- Assign to supervisor, set due date

API: POST /api/v1/capa-actions

Step 8: Update Investigation Status

- Mark investigation: In Progress → Completed

API: PATCH /api/v1/incidents/{id}

Conduct Safety Walk

Step 9: Safety Walk

- Select location/station
- Rate: housekeeping (1-5), compliance (1-5)
- Log issues found, critical issues count
- Mark follow-up required (Yes/No)
- Submit walk

API: `POST /api/v1/safety-walks`

Safety walks directly update: Dashboard compliance rating, Audit Readiness Score, Compliance Trend chart.

AUDITOR WORKFLOW

■ Auditor

Tabs:

Dashboard

Reports

Profile

■■ READ-ONLY — No create/update/delete permissions

Dashboard

Step 1: View Compliance Summary

- Leading indicators (TRIR, LTIFR, DART)
- Compliance trend chart
- Open CAPA count
- Active permits count

Reports (All Read-Only)

Step 2: View Incidents

- All incidents list with severity, date, location
 - Tap for full detail (read-only)
- API: `GET /api/v1/incidents`

Step 3: View Near Misses

- All near miss reports with potential consequence
- API: `GET /api/v1/near-misses`

Step 4: View Checklists

- All submitted/approved/rejected checklists
 - View item responses (cannot modify)
- API: `GET /api/v1/checklists/submissions`

Step 5: View CAPA Actions

- All CAPA actions with status, due date, assignee
- API: `GET /api/v1/capa-actions`

Step 6: View Safety Walks

- Historical inspection records
 - Compliance and housekeeping ratings
- API: `GET /api/v1/safety-walks`

Step 7: View Equipment Certifications

- All equipment certs with expiry dates
 - Overdue inspections flagged
- API: `GET /api/v1/equipment-certifications`

Auditor cannot: report incidents, submit checklists, approve permits, create CAPAs, or modify any data.

DATA FLOW: APK → BACKEND → WEB DASHBOARD

Action	API Call	Web Dashboard Impact
Worker reports incident	POST /incidents	Dashboard: Total Incidents +1, TRIR recalculates
Supervisor submits checklist	POST /checklists/.../submit	HSE Manager Checklist tab: new item for validation
HSE Manager validates checklist	POST /checklists/.../validate	Compliance metrics update, submission rate changes
HSE Manager approves permit	PATCH /permits-to-work/{id}	Dashboard: Active Permits count +1
HSE Manager creates CAPA	POST /capa-actions	Supervisor Tasks tab: new action item assigned
HSE Manager conducts safety walk	POST /safety-walks	Dashboard: Audit Readiness Score updates, Compliance Trend
Supervisor updates CAPA	PATCH /capa-actions/{id}	Dashboard: Open CAPA count changes
Worker reports near miss	POST /near-misses	Dashboard: Near Miss Ratio updates

PERMISSION MATRIX

Action	Worker	Supervisor	HSE Manager	Auditor
Report Incident	■	■	■	■
Report Near Miss	■	■	■	■
Submit Checklist	■	■	■	■
Validate Checklist	■	■	■	■
Approve/Create Permit	■	■	■	■
Conduct Safety Walk	■	■	■	■
Create CAPA Action	■	■	■	■
Update CAPA Status	■	■	■	■
View All Reports	■	■	■	■
View Training	■	■	■	■

Action	Worker	Supervisor	HSE Manager	Auditor
Perform Tasks	■	■	■	■